

Introduction to Python

ISA, Masters in Green Data Science, 2025-2026

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Guidelines for the final Project (November 2025)¹

Once you have solved each of the course's problem sets, it's time to implement your final project, a Python program of your very own! The design and implementation of your project is entirely up to you, albeit subject to the following requirements.

Requirements

1. Your project must be implemented in Python.
2. Your project must have a main function and three or more additional functions. At least three of those additional functions must be accompanied by tests that can be executed with pytest.
3. Your main function must be in a file called **project.py**, which should be in the "root" (i.e., top-level folder) of your project.
4. Your 3 required custom functions other than **main()** must also be in **project.py** and defined at the same indentation level as **main** (i.e., not nested under any classes or functions).
5. Your test functions must be in a file called **test_project.py**, which should also be in the "root" of your project. Be sure they have the same name as your custom functions, prepended with **test_** (**test_custom_function**, for example, where **custom_function** is a function you've implemented in **project.py**).
6. You should implement at least one class, which should have at least one user defined method.
7. You are welcome to implement additional classes and functions as you see fit beyond the minimum requirement.
8. Implementing your project should entail more time and effort than is required by each of the course's problem sets.
9. Avoid or limit manual inputs from the user: inputs should be in general included in the code, or read from files.
10. Any pip-installable libraries that your project requires must be listed, one per line, in a file called **requirements.txt** in the root of your project (see example below).

Example of **project.py**

```
def main():
```

¹ Loosely adapted from <https://cs50.harvard.edu/python/2022/project/>

```
...
def function_1():
...
def function_2():
...
def function_n():
...
if __name__ == "__main__":
    main()
```

Example of test_project.py

```
def test_function_1():
...
def test_function_2():
...
def test_function_n():
...
```

Example of file requirements.txt:

```
pandas
numpy
re
csv
sys
...
```

You are welcome, but not required, to collaborate with one or two classmates on your project. But a two- or three-person should entail twice or thrice the time and effort required by a one-person project.

Getting Started

Creating an entire project may seem daunting. Here are some questions that you should think about as you start:

What will your software do? What features will it have? How will it be executed?

What new skills will you need to acquire? What topics will you need to research?

If working with one or two classmates, who will do what?

In the world of software, most everything takes longer to implement than you expect. And so it's not uncommon to accomplish less in a fixed amount of time than you hope. What might you consider to be a good outcome for your project? A better outcome? The best outcome?

Consider making goal milestones to keep you on track.

How to Submit

See guidelines on Moodle.

Evaluation guidelines

There will be a discussion with the instructor where all elements of the group should be present. The grade depends on the group project but it is individual (i.e. distinct element of the group might get distinct grades).

Evaluation criteria

- Adherence to the requirements (see above)
- Clarity of the report (README.md in repository) and video. Low: if the report is poor and doesn't explain clearly what the data and problem are. High: if data and problem are well described, if the approach to solve the problem is carefully justified, and if the video presentation is engaging.
- Relevance and novelty of the project. Low: if it is very similar to what was done in class or too simple. High: if the project is original, compelling and more complex than the problems discussed in class.
- Technical quality. Low: if the code to solve the problem is unnecessarily complex or if it contains errors. High: if it's correct and uses judiciously the techniques discussed in class to solve the problem at hand.
- Code organization (modularity, comments). Low: if the script is long, involved and difficult to follow. High: if there is a compact and clear main() function that establishes the main steps of the resolution, and calls several additional functions, each one commented and organized in a way that the code is clear and easy to follow and modify if needed.
- Testing. Low: if the test functions do not cover obvious interesting 'corner' cases. High: if test functions are sound and reliable.

To get a 'pass', the following requirements are necessary:

- After cloning the repository and installing the required packages, 'project.py' should be executed with no execution errors, creating the expected output(s)
- The project needs to be different from what was done in class, but can be a variation on a problem already discussed in class, with some changes in the scripts, inputs and outputs. This should be described in the report, where the changes with respect to what was done in class should be discussed.
- You should implement at least one class, which should have at least one user defined method.
- Test functions need to be included in the project.

Some past project topics:

- Python weather analyser: This project uses a user's IP address to obtain historical weather based on the area that they are in, using the Open-Meteo weather API (<https://github.com/eleshuk/python-final-project/>)
- Scrabble Training: We created the Scrabble Training program that enhances the user's ability to find the best words from random letters through interactive practice sessions. The program has three practice modes to help users improve their performance in the game, focusing on identifying the longest words and the highest scoring words that can be made from the provided letters (https://github.com/domwelsh/scrabble_training)

- Tool for customized and fast downloading of satellite images (https://github.com/Emmanuel461/Final_project)
- Pest Tracker tem como objetivo simular uma aplicação para a previsão da praga da mosca da azeitona no olival (<https://github.com/filipefelisardo/PestTracker>)
- Forecast Hunter: A toolbox for assessing seasonal weather forecast data usability on ETo estimation (<https://github.com/dgarcian9/ForecastHunter>)
- Determine fertilization depending on culture parameters and soil composition analysis (<https://github.com/RubenRomeroTorrado/Fertilizacao/>)
- Pokemon_battle_simulator: This project consists on a pokemon battle simulator made with Python. It uses an graphical user interface (GUI) to display the battles, as in the original pokemon games (https://github.com/nachiet/Pokemon_battle_simulator)
- Weather report: code to provide the user with a weather report for almost any city in the world in any date of choice (<https://github.com/marianadc01/ipython-weather-report/>)
- Calculation of Urban Density with QGIS and Python (<https://github.com/laurafi97/submission-python-qgis->)
- Word Wizardry (https://github.com/Sofs27/Final_Project_Alicia_Sofia_1semestre)